Mathematics Mock Exam

Number sense, arithmetic, geometry, and problem solving.

70 original multiple-choice practice questions ? Choose the best answer for each item.

Unofficial practice material created for ScholarHunt. This is not an official scholarship examination.

1. What is 18 + 27?

A. 35
B. 45
C. 46
D. 55

2. What is 84 − 39?

A. 35
B. 45
C. 55
D. 65

3. What is 7 × 8?

A. 48
B. 54
C. 56
D. 64

4. What is 96 ÷ 12?

A. 6
B. 7
C. 8
D. 9

5. What is 3/4 of 20?

A. 5
B. 10
C. 15
D. 16

6. Which fraction is equal to 1/2?

A. 2/3
B. 3/6
C. 4/6
D. 5/8

7. What is 0.6 + 0.25?

A. 0.31
B. 0.75

C. 0.85
D. 0.95

8. What is 25% of 80?

A. 10
B. 15
C. 20
D. 25

9. Simplify the ratio 15:20.

A. 2:3
B. 3:4
C. 4:5
D. 5:4

10. What is the perimeter of a square with side 6 cm?

A. 12 cm
B. 18 cm
C. 24 cm
D. 36 cm

11. What is the area of a rectangle 8 cm long and 3 cm wide?

A. 11 cm^2
B. 22 cm^2
C. 24 cm^2
D. 48 cm^2

12. What is the value of 5^2?

A. 10
B. 15
C. 20
D. 25

13. Which number is prime?

A. 21
B. 27
C. 29
D. 33

14. What is the least common multiple of 4 and 6?

A. 8
B. 10
C. 12
D. 24

15. What is the greatest common factor of 18 and 24?

A. 3

B. 6
C. 9
D. 12

16. Solve: x + 9 = 17.

A. 6
B. 7
C. 8
D. 9

17. Solve: 4x = 28.

A. 6
B. 7
C. 8
D. 9

18. What is the value of 3 + 2 × 5?

A. 13
B. 25
C. 17
D. 30

19. Round 7,462 to the nearest hundred.

A. 7,400
B. 7,450
C. 7,500
D. 8,000

20. What is 2.4 × 3?

A. 6.2
B. 7.2
C. 7.4
D. 8.2

21. What is 9.6 ÷ 3?

A. 2.2
B. 3.2
C. 3.6
D. 4.2

22. A shirt costs ₱500 and is discounted by 10%. What is the sale price?

A. ₱50
B. ₱400
C. ₱450
D. ₱490

23. What is the mean of 4, 6, and 8?

A. 5
B. 6
C. 7
D. 18

24. What is the median of 2, 7, 9, 12, 15?

A. 7
B. 9
C. 10
D. 12

25. A fair coin is tossed once. What is the chance of heads?

A. 0
B. 1/4
C. 1/2
D. 1

26. How many degrees are in a right angle?

A. 45
B. 90
C. 180
D. 360

27. How many sides does a hexagon have?

A. 5
B. 6
C. 7
D. 8

28. What is the volume of a cube with side length 3 cm?

A. 9 cm^3
B. 18 cm^3
C. 27 cm^3
D. 36 cm^3

29. Convert 2.5 meters to centimeters.

A. 25 cm
B. 250 cm
C. 2,500 cm
D. 0.25 cm

30. How many minutes are in 2 hours?

A. 60
B. 100
C. 120
D. 200

31. What is 1/3 + 1/3?

A. 1/6
B. 1/3
C. 2/3
D. 1

32. What is 5/6 − 1/6?

A. 1/6
B. 2/3
C. 4/5
D. 1

33. Which decimal is greatest?

A. 0.45
B. 0.5
C. 0.49
D. 0.405

34. What is the next multiple of 7 after 35?

A. 40
B. 42
C. 45
D. 49

35. If 3 notebooks cost ₱90, what do 5 cost at the same rate?

A. ₱120
B. ₱135
C. ₱150
D. ₱180

36. What is 15% of 200?

A. 15
B. 20
C. 30
D. 40

37. What is 1.2 + 0.08?

A. 0.20
B. 1.28
C. 1.80
D. 2.00

38. What is 7.5 − 2.35?

A. 4.15
B. 5.05
C. 5.15
D. 6.15

39. Which is an equivalent ratio to 2:5?

A. 4:10
B. 4:5
C. 5:2
D. 6:10

40. A car travels 120 km in 2 hours. What is its average speed?

A. 40 km/h
B. 50 km/h
C. 60 km/h
D. 240 km/h

41. What is the area of a triangle with base 10 cm and height 4 cm?

A. 14 cm^2
B. 20 cm^2
C. 40 cm^2
D. 80 cm^2

42. A circle has radius 5 cm. What is its diameter?

A. 2.5 cm
B. 5 cm
C. 10 cm
D. 25 cm

43. What is the sum of the angles in a triangle?

A. 90 degrees
B. 180 degrees
C. 270 degrees
D. 360 degrees

44. What is the value of 2^3?

A. 6
B. 8
C. 9
D. 12

45. Solve: $3x + 2 = 14$.

A. 3
B. 4
C. 5
D. 6

46. If $y = 5$, what is $2y + 3$?

A. 10
B. 11
C. 13
D. 15

47. What is the absolute value of −9?

A. −9
B. 0
C. 9
D. 18

48. Which integer is greatest?

A. −8
B. −3
C. −1
D. −5

49. What is 40% expressed as a fraction in simplest form?

A. 1/4
B. 2/5
C. 4/5
D. 4/10

50. A class has 12 girls and 18 boys. What fraction of the class are girls?

A. 2/5
B. 3/5
C. 2/3
D. 1/3

51. What is the perimeter of a rectangle 9 m by 4 m?

A. 13 m
B. 26 m
C. 36 m
D. 72 m

52. How many faces does a cube have?

A. 4
B. 6
C. 8
D. 12

53. What is 3,600 seconds equal to?

A. 30 minutes
B. 45 minutes
C. 1 hour
D. 2 hours

54. What is 4 kilograms in grams?

A. 40 g
B. 400 g
C. 4,000 g
D. 40,000 g

55. What is the probability of rolling a 3 on a fair six-sided die?

A. 1/2
B. 1/3
C. 1/6
D. 3/6

56. A bag has 3 blue and 7 red marbles. What is the probability of blue?

A. 3/7
B. 3/10
C. 7/10
D. 1/3

57. What is the range of 4, 9, 11, and 16?

A. 7
B. 11
C. 12
D. 20

58. What is the mode of 2, 3, 3, 4, 5?

A. 2
B. 3
C. 4
D. 5

59. Which number is divisible by 3?

A. 124
B. 231
C. 415
D. 502

60. What is 11 × 12?

A. 121
B. 132
C. 144
D. 156

61. What is 144 ÷ 9?

A. 14
B. 15
C. 16
D. 18

62. A recipe needs 2 cups of flour for 8 servings. How much for 4 servings?

A. 1/2 cup
B. 1 cup
C. 2 cups
D. 4 cups

63. A map scale is 1 cm to 5 km. What distance does 4 cm represent?

A. 9 km
B. 20 km
C. 25 km
D. 45 km

64. What is the complement of a 35-degree angle?

A. 35 degrees
B. 55 degrees
C. 65 degrees
D. 145 degrees

65. What is the supplement of a 120-degree angle?

A. 30 degrees
B. 60 degrees
C. 90 degrees
D. 240 degrees

66. Which point lies on the x-axis?

A. (0, 4)
B. (3, 0)
C. (2, 5)
D. (−1, 2)

67. What is the next number in 2, 5, 8, 11, ...?

A. 12
B. 13
C. 14
D. 15

68. What is 5/8 expressed as a decimal?

A. 0.58
B. 0.625
C. 0.75
D. 0.8

69. A ₱1,000 item has a 20% discount. What is the discounted price?

A. ₱200
B. ₱700
C. ₱800
D. ₱980

70. A 3-meter ribbon is cut into pieces of 25 centimeters each. How many pieces are made?

A. 8
B. 10
C. 12
D. 15

Answer Key & Explanations

Use this section after completing the practice exam.

1. B ? 45
Adding the ones and tens gives 45.

2. B ? 45
84 minus 39 equals 45.

3. C ? 56
Seven groups of eight equal 56.

4. C ? 8
Twelve times eight equals 96.

5. C ? 15
One-fourth of 20 is 5, so three-fourths is 15.

6. B ? 3/6
Three-sixths simplifies to one-half.

7. C ? 0.85
Six tenths plus twenty-five hundredths is 0.85.

8. C ? 20
One-fourth of 80 is 20.

9. B ? 3:4
Divide both terms by five to get 3:4.

10. C ? 24 cm
A square has four equal sides; 4 × 6 = 24.

11. C ? 24 cm²
Area is length times width: 8 × 3 = 24.

12. D ? 25
Five squared means 5 × 5.

13. C ? 29
29 has no positive factors other than 1 and itself.

14. C ? 12
Twelve is the smallest number divisible by both 4 and 6.

15. B ? 6
Six is the greatest number that divides both evenly.

16. C ? 8
Subtract nine from both sides to get x = 8.

17. B ? 7
Divide both sides by four.

18. A ? 13
Multiply first: 3 + 10 = 13.

19. C ? 7,500
The tens digit is six, so round the hundreds up.

20. B ? 7.2
Three groups of 2.4 total 7.2.

21. B ? 3.2
Three times 3.2 equals 9.6.

22. C ? ₱450
The discount is ₱50; subtracting it leaves ₱450.

23. B ? 6
Their sum is 18, and 18 divided by 3 is 6.

24. B ? 9
The middle value in the ordered list is 9.

25. C ? 1/2
There are two equally likely outcomes and one is heads.

26. B ? 90
A right angle measures 90 degrees.

27. B ? 6
The prefix hexa- means six.

28. C ? 27 cm³
Volume is side cubed: 3 × 3 × 3 = 27.

29. B ? 250 cm
One meter is 100 centimeters, so 2.5 m is 250 cm.

30. C ? 120
Each hour has 60 minutes; 2 × 60 = 120.

31. C ? 2/3
The denominators match, so add the numerators.

32. B ? 2/3
Four-sixths simplifies to two-thirds.

33. B ? 0.5
0.5 is five tenths, greater than each listed value.

34. B ? 42
Seven times six is 42.

35. C ? ₱150
Each notebook costs ₱30, so five cost ₱150.

36. C ? 30
Ten percent is 20 and five percent is 10, totaling 30.

37. B ? 1.28
Twelve tenths plus eight hundredths is 1.28.

38. C ? 5.15
Aligning decimal places gives 7.50 − 2.35 = 5.15.

39. A ? 4:10
Multiplying both ratio terms by two gives 4:10.

40. C ? 60 km/h
Average speed is distance divided by time: 120 ÷ 2 = 60.

41. B ? 20 cm²
Triangle area is one-half times base times height.

42. C ? 10 cm
The diameter is twice the radius.

43. B ? 180 degrees
A triangle’s interior angles total 180 degrees.

44. B ? 8
Two cubed is 2 × 2 × 2 = 8.

45. B ? 4
Subtract 2, then divide 12 by 3.

46. C ? 13
Substitute 5: 2 × 5 + 3 = 13.

47. C ? 9
Absolute value is distance from zero, so it is nonnegative.

48. C ? −1
Among negative numbers, the one closest to zero is greatest.

49. B ? 2/5
Forty percent is 40/100, which reduces to 2/5.

50. A ? 2/5
There are 30 students; 12/30 simplifies to 2/5.

51. B ? 26 m
Perimeter is 2 × (9 + 4) = 26 meters.

52. B ? 6
A cube has six square faces.

53. C ? 1 hour
There are 60 seconds per minute and 60 minutes per hour.

54. C ? 4,000 g
One kilogram equals 1,000 grams.

55. C ? 1/6
One of the six equally likely faces shows 3.

56. B ? 3/10
Three of the ten marbles are blue.

57. C ? 12
Range is the greatest value minus the least: 16 − 4 = 12.

58. B ? 3
Three appears more often than any other value.

59. B ? 231
The digits of 231 sum to 6, divisible by 3.

60. B ? 132
Eleven groups of twelve equal 132.

61. C ? 16
Nine times sixteen equals 144.

62. B ? 1 cup
Four servings are half as many, so use half the flour.

63. B ? 20 km
Four centimeters represent 4 × 5 = 20 kilometers.

64. B ? 55 degrees
Complementary angles total 90 degrees; 90 − 35 = 55.

65. B ? 60 degrees
Supplementary angles total 180 degrees; 180 − 120 = 60.

66. B ? (3, 0)
Points on the x-axis have a y-coordinate of zero.

67. C ? 14
The sequence adds three each time.

68. B ? 0.625
Five divided by eight equals 0.625.

69. C ? ₱800
Twenty percent is ₱200; subtracting gives ₱800.

70. C ? 12
Three meters is 300 centimeters; 300 divided by 25 is 12.